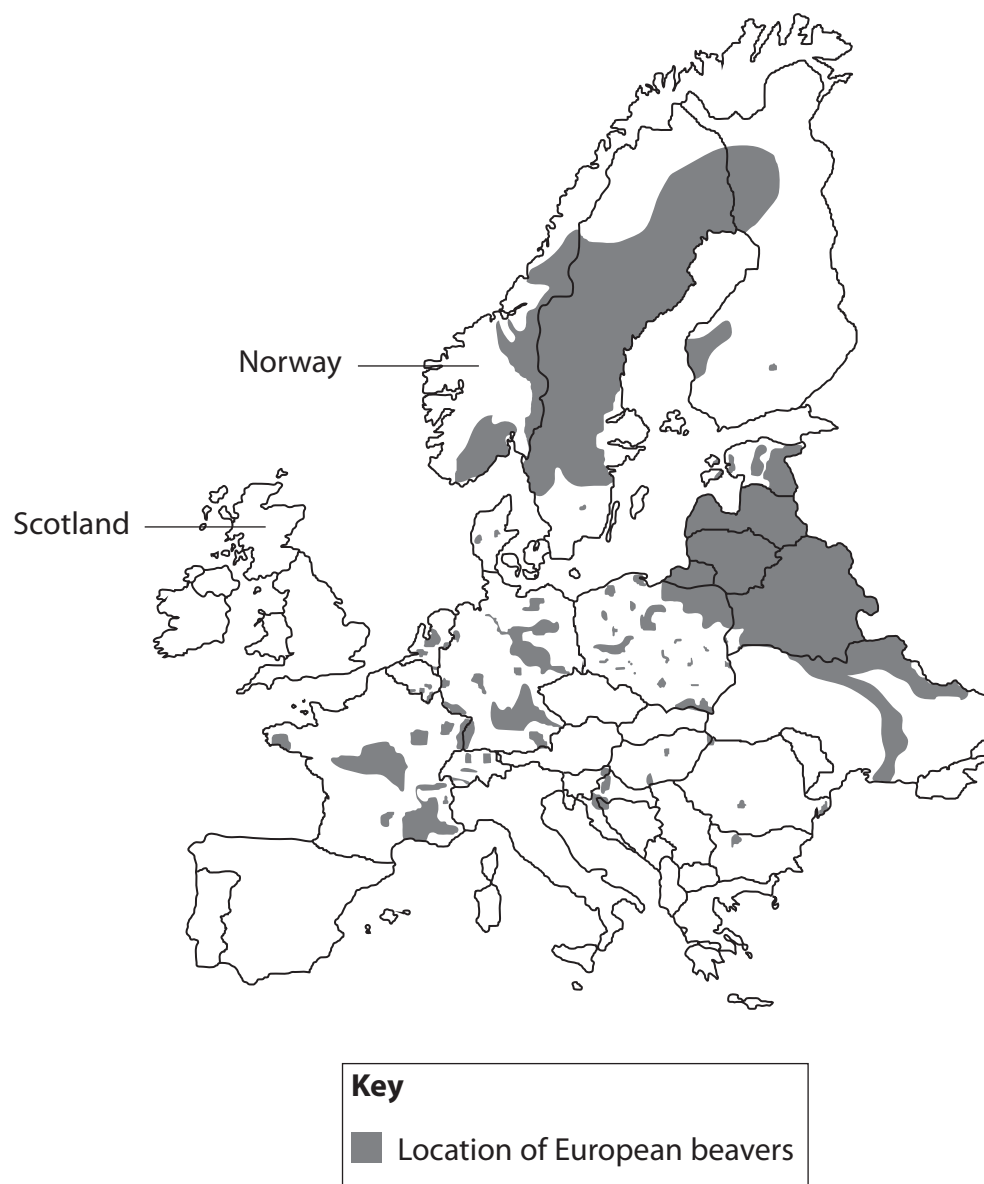


*(b) Some scientists were concerned that the low numbers of beavers reintroduced into Scotland from Norway could affect the genetic diversity of the population in Scotland.

The scientists proposed two solutions to this concern:

- to introduce more beavers from Norway
- to introduce beavers from several different European locations.

The map shows locations of European beavers before they were reintroduced to Scotland.



Discuss the solutions, proposed by these scientists, to overcome the concern of introducing only 11 beavers into Scotland from Norway.

Use the information in the map to support your answer.

(6)

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- (c) In a beaver population, the frequency of a recessive homozygous genotype is 0.09.

Calculate the percentage of beavers in this population that are homozygous for the dominant allele, using the equation

$$p^2 + 2pq + q^2 = 1$$

(3)

Answer%

(Total for Question 8 = 13 marks)

TOTAL FOR PAPER = 80 MARKS

